# Learning What to Learn: Adaptive Sample Selection for Efficient Neural Network Training

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## Abstract

Training neural networks on large datasets is increasingly expensive. Not all training samples contribute equally to learning at every stage of training. We investigate whether *adaptive sample selection*— choosing which examples to train on based on per-sample difficulty, uncertainty, and feature-space diversity— can improve training efficiency compared to uniform random sampling under a fixed data budget.

We propose a lightweight, forward-pass-only adaptive selector that combines cross-entropy loss, softmax entropy, and feature-space diversity with a phase-dependent mixing schedule. We implement four baselines (random, difficulty-only, uncertainty-only, diversity-only) and evaluate all methods on Fashion-MNIST [Xiao et al., 2017] using a compact CNN under data budgets of 10%–100% of the training pool, with three random seeds per configuration.

Our experiments suggest **[TBD: results pending E001–E005]**. We further present an ablation study isolating the contribution of each selection signal, and a failure analysis characterising conditions under which adaptive selection degrades relative to random sampling.

All code, configs, and raw results are publicly available at `https://github.com/[username]/adaptive-sample-selection`.

## 1 Introduction

Modern neural network training can be computationally expensive, even for moderate-scale datasets. A standard training run processes every training example once per epoch, treating all samples as equally informative. However, empirical evidence suggests this is not the case: some examples are learned quickly and never forgotten, while others remain persistently difficult throughout training [Toneva et al., 2019].

This raises a natural question: *can we select which examples to include in each training step, and in what order, to make learning more efficient?*

A long line of work addresses related problems. Curriculum learning [Bengio et al., 2009] orders examples by pre-defined difficulty. Self-paced learning [Kumar et al., 2010] determines difficulty dynamically from model loss. Active learning [Ren et al., 2021] selects which examples to *label*; in our setting, labels are available for all data. Coreset selection [Mirzasoleiman et al., 2020] finds representative subsets; our approach is lighter-weight and operates online.

**Research question.** Can lightweight adaptive sample selection improve training efficiency— measured as validation accuracy per unit of data seen—compared to uniform random sampling, when all methods operate under the same fixed data budget?

**Contributions.** We make the following contributions:

1. A modular, reproducible framework for comparing sample-selection strategies under fixed data budgets.
2. A controlled empirical study on Fashion-MNIST across five methods, five budget levels, and three random seeds.
3. A lightweight adaptive selector combining difficulty, uncertainty, and diversity with a phase-dependent schedule.
4. An ablation study and failure analysis documenting when adaptive selection helps and when it does not.

We do not claim algorithmic novelty over the existing literature. Our contribution is primarily *empirical*: a rigorous, reproducible comparison under

a CPU-scale setting that is underrepresented in the literature.

## 2 Related Work

**Curriculum learning.** Bengio et al. [2009] showed that presenting examples from easy to hard improves convergence and generalisation. The curriculum is pre-defined and not adaptive to the model state.

**Self-paced learning.** Kumar et al. [2010] introduced SPL, which jointly learns model parameters and example selection weights using a regularised objective. Jiang et al. [2015] extended this with a predefined curriculum prior (SPCL). Jiang et al. [2014] added a diversity constraint (SPLD), directly motivating our proposed method.

**Hard example mining.** Shrivastava et al. [2016] selects highest-loss samples per batch in detection (OHEM). Lin et al. [2017] achieves similar effects through a differentiable loss reweighting (Focal Loss). Both focus on difficulty without diversity.

**Coreset selection.** Mirzasoleiman et al. [2020] treats subset selection as submodular optimisation, approximating the full gradient with a small representative set. This is powerful but requires gradient computation over the full pool— expensive for CPU-only settings.

**Training dynamics.** Toneva et al. [2019] showed that many training examples are redundant: once learned, they are never forgotten and can be removed without accuracy loss. This motivates our difficulty/loss-based scoring.

**Data valuation.** Ghorbani and Zou [2019] use Shapley values to assess individual data point importance, but exact computation is prohibitive. We use tractable proxies: loss and entropy.

**Data pruning.** Sorscher et al. [2022] demonstrate that selecting harder, less-central examples outperforms random pruning at small data fractions— directly relevant to our small-budget experiments.

**Importance sampling.** Zhao and Zhang [2015] show that non-uniform sampling proportional to gradient norm yields provably faster convergence, providing theoretical justification for our loss-based scoring.

Our work differs from active learning [Settles, 2009] in that all labels are available; we select for training efficiency, not annotation efficiency.

## 3 Problem Formulation

Let $\mathcal{D} = \{(x_i, y_i)\}_{i=1}^{N}$ be the full training pool, and let $f_\theta$ be a neural network with parameters $\theta$. We define a *data budget* $B \in (0, 1]$ such that each epoch selects $n = \lfloor B \cdot N \rfloor$ samples from $\mathcal{D}$.

A *selector* $\pi : \mathcal{D} \times f_\theta \times t \to \mathcal{I}$ maps the current model state and training time $t$ to a set of $n$ indices $\mathcal{I} \subset [N]$, $|\mathcal{I}| = n$.

The training objective is:

$$\theta^* = \arg\min_\theta \mathbb{E}_{i\sim\pi}[\ell(f_\theta(x_i), y_i)] \tag{1}$$

where $\ell$ is cross-entropy loss.

**Efficiency metric.** Our primary measure is validation accuracy $a(s)$ as a function of cumulative training samples seen $s$. A method $\pi$ is more efficient than random baseline $\pi_0$ if, for some target accuracy $a^*$:

$$s_\pi(a^*) < s_{\pi_0}(a^*) \tag{2}$$

where $s_\pi(a^*)$ is the number of samples seen to first achieve $a^*$.

## 4 Method

### 4.1 Sample Scoring

For each sample $x_i$ at epoch $t$, we compute three scores using only forward passes (no gradient computation beyond the training update):

**Difficulty $D_i(t)$.** The normalised per-sample cross-entropy loss:

$$D_i(t) = \frac{\ell_i(t) - \min_j \ell_j(t)}{\max_j \ell_j(t) - \min_j \ell_j(t) + \epsilon} \tag{3}$$

where $\ell_i(t) = -\log p_\theta(y_i|x_i)$ at epoch $t$. High $D_i$ indicates the model currently finds $x_i$ difficult.

**Uncertainty $U_i(t)$.** The normalised softmax entropy:

$$U_i(t) = \frac{H_i(t) - \min_j H_j(t)}{\max_j H_j(t) - \min_j H_j(t) + \epsilon} \tag{4}$$

where $H_i(t) = -\sum_c p_\theta(c|x_i) \log p_\theta(c|x_i)$. High $U_i$ indicates the model is uncertain about the class of $x_i$.

**Diversity $V_i(t)$.** The normalised minimum distance from sample $x_i$'s feature representation to the nearest previously-selected centroid:

$$V_i(t) = \frac{d_i(t) - \min_j d_j(t)}{\max_j d_j(t) - \min_j d_j(t) + \epsilon} \tag{5}$$

where $d_i(t) = \min_{c \in C_{t-1}} \|\phi(x_i) - c\|_2$, $\phi(x_i)$ is the penultimate-layer feature, and $C_{t-1}$ is the set of selected sample features from the previous epoch.

### 4.2 Adaptive Selection Policy

The combined score for sample $x_i$ at epoch $t$ is:

$$S(x_i, t) = \alpha(t) \cdot D_i(t) + \beta(t) \cdot U_i(t) + \gamma(t) \cdot V_i(t) \tag{6}$$

where $\alpha(t), \beta(t), \gamma(t) \geq 0$ are phase-dependent weights with $\alpha + \beta + \gamma = 1$.

**Phase schedule.** We use a three-phase schedule motivated by curriculum learning:

$$\text{Phase 1 (epochs 1–10):} \quad (\alpha, \beta, \gamma) = (0.2, 0.2, 0.6) \tag{7}$$

$$\text{Phase 2 (epochs 11–20):} \quad (\alpha, \beta, \gamma) = (0.4, 0.4, 0.2) \tag{8}$$

$$\text{Phase 3 (epochs 21+):} \quad (\alpha, \beta, \gamma) = (0.5, 0.4, 0.1) \tag{9}$$

Early training emphasises diversity (broad exploration of the feature space). Late training emphasises difficulty (focus on boundary-clarifying examples). Ablation experiments (Section 7) validate this schedule.

**Selection rule.** Select the top-$n$ samples by score $S(x_i, t)$:

$$\mathcal{I}_t = \arg\operatorname*{top}_n \{S(x_i, t) : i \in [N]\} \tag{10}$$

**Computational cost.** One forward pass over the pool: $\mathcal{O}(N)$. Sorting: $\mathcal{O}(N \log N)$. Diversity scoring (over candidate subset): $\mathcal{O}(n \cdot |\mathcal{C}|)$. Memory: $\mathcal{O}(N \cdot D)$ for the feature matrix.

Setting score_every $= 1$ recomputes scores each epoch (highest quality, highest overhead). Setting score_every $= k$ reduces overhead by $k\times$ with a slight staleness penalty.

### 4.3 Baselines

We compare against four baselines:

1. **Random** ($\pi_{\text{rand}}$): uniform sampling without replacement.
2. **Difficulty** ($\pi_D$): top-$n$ by $D_i(t)$ alone.
3. **Uncertainty** ($\pi_U$): top-$n$ by $U_i(t)$ alone.
4. **Diversity** ($\pi_V$): k-means clustering; select nearest-to-centroid from each cluster.

## 5 Experimental Setup

### 5.1 Dataset

We use Fashion-MNIST [Xiao et al., 2017]: 60,000 training images and 10,000 test images, 28×28 grayscale, 10 balanced classes. We split training into a pool of 50,000 and a validation set of 10,000 (stratified by class). The test set is used only for final evaluation.

### 5.2 Model

Our primary architecture (SmallCNN) is a compact CNN with two convolutional blocks (32 and 64 channels) followed by a fully-connected classifier ( 450k parameters). A secondary MLP ( 235k parameters) is used for robustness experiments (Section **[TBD: E010]**).

### 5.3 Training Protocol

All methods use identical training protocols:

- Optimiser: Adam (lr $= 10^{-3}$, weight decay $10^{-4}$)
- Batch size: 128
- Epochs: 30
- Data augmentation: random horizontal flip (training only)

### 5.4 Data Budgets

Budget $B \in \{0.10, 0.25, 0.50, 0.75, 1.00\}$. At budget $B$, each epoch selects $n = B \times 50\,000$ samples. All methods receive exactly $n$ samples per epoch.

### 5.5 Evaluation

Three random seeds per configuration (42, 123, 777). We report mean ± standard deviation across seeds. Primary metric: validation accuracy at equal cumulative samples seen. Secondary metrics: macro F1, wall-clock time, gradient updates.

### 5.6 Hardware

All experiments run on a CPU-only Intel laptop (8 cores, 8.3 GB RAM). No GPU is used, ensuring results are reproducible without specialist hardware.

Table 1: Validation accuracy (mean ± std, 3 seeds) by method and data budget.

| Method | 10% | 25% | 50% | 100% |
|---|---|---|---|---|
| Random | [TBD: ] | [TBD: ] | [TBD: ] | [TBD: ] |
| Difficulty | [TBD: ] | [TBD: ] | [TBD: ] | [TBD: ] |
| Uncertainty | [TBD: ] | [TBD: ] | [TBD: ] | [TBD: ] |
| Diversity | [TBD: ] | [TBD: ] | [TBD: ] | [TBD: ] |
| Adaptive | [TBD: ] | [TBD: ] | [TBD: ] | [TBD: ] |

# 6 Results

**[TBD: Results pending completion of experiments E001–E005. Table 1 and Figures ??–?? will be populated from results/processed/mean_std.csv after all runs complete.]**

# 7 Ablation Study

**[TBD: Ablation results pending E006. Will compare all 7 component combinations (D only, U only, V only, D+U, D+V, U+V, D+U+V) at budget 25%.]**

# 8 Failure Analysis

**[TBD: Failure analysis pending E007–E009. Will report performance under: (a) 10% and 20% label noise, (b) 5:1 class imbalance, (c) 30% duplicate data.]**

**Anticipated failure modes.** Based on theory and related work, we anticipate the following:

1. **Label noise amplification**: The difficulty selector is expected to over-select mislabelled examples (high loss $\not\Rightarrow$ informative). The adaptive selector's diversity component may partially mitigate this.

2. **Class imbalance**: A purely difficulty-based selector may over-select the minority class (harder to classify). Diversity mitigates this by enforcing coverage across feature clusters.

3. **Redundant data**: The difficulty selector may repeatedly select the same hard examples, wasting the budget on redundant samples.

# 9 Discussion

**[TBD: Discussion pending experimental results.]**

**If H1 is supported (adaptive > random at small budgets):** This would confirm that lightweight, forward-pass-only selection provides a meaningful efficiency advantage over random sampling on Fashion-MNIST.

**If H2 is supported (difficulty hurt at 10% budget):** This would be a useful negative result, warning practitioners against naive hard-example mining when the training pool may contain noisy examples and when diversity is critical at very small budgets.

**If H0 is not rejected (no significant advantage):** This would be a valid scientific result suggesting that for small, balanced datasets like Fashion-MNIST, adaptive selection overhead does not justify the gains, and random sampling should be preferred for simplicity.

# 10 Limitations

1. **Single dataset**: All primary experiments use Fashion-MNIST. Results may not generalise to larger datasets or more complex vision tasks.

2. **CPU-only training**: Wall-clock time comparisons are specific to our hardware. GPU experiments may show different overhead characteristics.

3. **Three seeds**: Statistical power is limited. We report mean ± std but acknowledge that larger-scale replication is desirable.

4. **Hyperparameter sensitivity**: The phase schedule weights $(\alpha, \beta, \gamma)$ are fixed from preliminary ablation. They may not be optimal for other datasets or architectures.

5. **Tractable proxies**: We use loss and entropy as proxies for true data importance. Shapley values or gradient-based methods may provide better estimates at higher computational cost.

# 11 Conclusion

**[TBD: Conclusion pending experimental results.]**

We present a controlled empirical study of lightweight adaptive sample selection for efficient neural network training under fixed data budgets. Our proposed method combines per-sample difficulty, uncertainty, and feature-space diversity signals with a phase-dependent schedule, motivated by the Self-Paced Learning with Diversity framework [Jiang et al., 2014] and curriculum learning principles [Bengio et al., 2009].

The primary findings of this study will determine whether **[TBD: H1 or H0]**.

Regardless of outcome, this work establishes a reusable framework and a reproducible baseline for CPU-scale adaptive sample selection research— a contribution accessible to undergraduate researchers without specialist hardware.

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